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感谢您的关注，产品详情如下：



EV5357/EV5357M_S0_R0

18V/3A Synchronous Step-Down Converter

Parameters Subject to Change Without Notice

FEATURES

- 4.5V to 18V operating input range
3A output current
- Up to 95% efficiency
- PFM at light load
- 600kHz switching frequency
- Internal soft-start
- Input under-voltage lockout
- Current run-away protection
- Output short protection
- Thermal protection
- Available in SOT23-6 package

APPLICATIONS

- Distributed Power Systems
- Networking Systems
- FPGA, DSP, ASIC Power Supplies
- Green Electronics/ Appliances
- Notebook Computers

DESCRIPTION

The JW[®]5357 and JW[®]5357M are monolithic buck switching regulator based on I2 architecture for fast transient response. Operating with an input range of 4.5V~18V, JW5357 and JW5357M deliver 3A of continuous output current with two integrated N-Channel MOSFETs. The internal synchronous power switches provide high efficiency without the use of an external Schottky diode. At light loads, JW5357 and JW5357M operate in low frequency to maintain high efficiency.

JW5357 and JW5357M guarantee robustness with output short protection, thermal protection, current run-away protection and input under voltage lockout.

JW5357 and JW5357M are available in SOT23-6 package, which provide a compact solution with minimal external components.

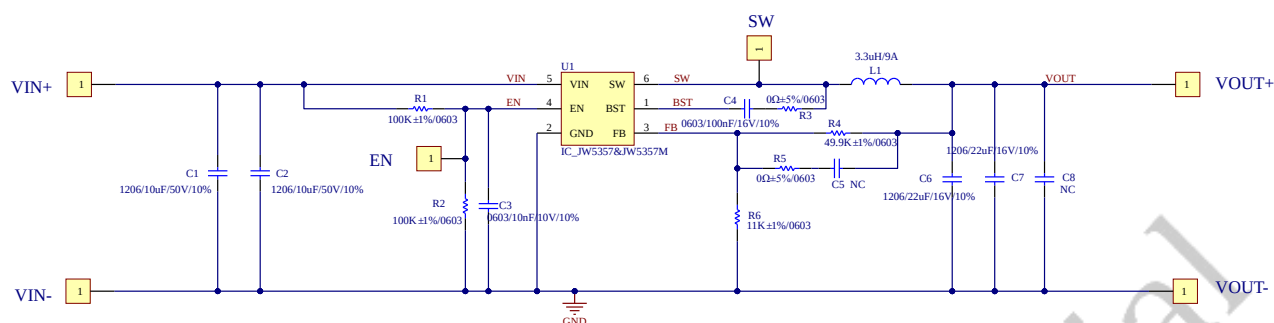
ELECTRICAL SPECIFICATIONS

Parameter	Symbol	Value	Unit
Input Voltage	VIN	4.5~18	V
Output Voltage	VOUT	3.3	V
Output Current	IOUT	0~3	A

EVALUATION BOARD AND TYPICAL PERFORMANCE

EV5357&EV5357M 60mm×50mm

SCHEMATIC



BILL OF MATERIALS

Qty	Designator	Value	Description	Package	Manufacturer	Manufacturer P/N
2	C1,C2	10uF	Ceramic capacitor 50V, X7R	1206C		
1	C3	10nF	Ceramic capacitor 10V, X7R	0603C		
1	C4	100nF	Ceramic capacitor 16V, X7R	0603C		
1	C5	NC				
2	C6,C7	22uF	Ceramic capacitor 16V, X7R	1206C		
1	L1	3.3uH/9A	Inductor	L12*12	WE	744314330
2	R1, R2	100k	Resistor,1%	0603R		
2	R3, R5	0 Ω	Resistor,5%	0603R		
1	R4	49.9k	Resistor,1%	0603R		
1	R6	11K	Resistor,1%	0603R		
1	VIN	4.5V~18V		TEST-Pole		
1	VOUT	3.3V/3A		TEST-Pole		
1	JW5357/JW5357M	18V/3A	Buck	SOT23-6	Joulwatt	JW5357/JW5357M

PRINTED CIRCUIT BOARD LAYEROUT

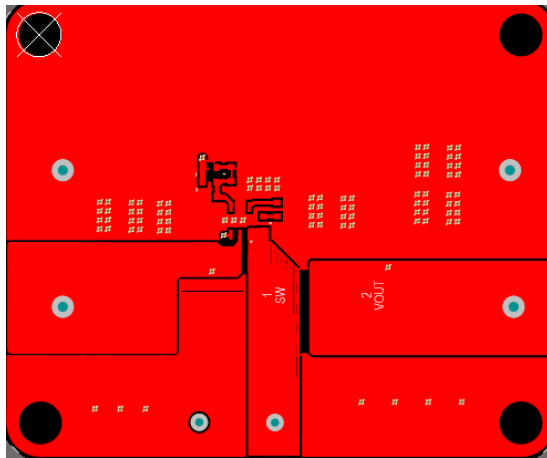


Figure1—Top Layer

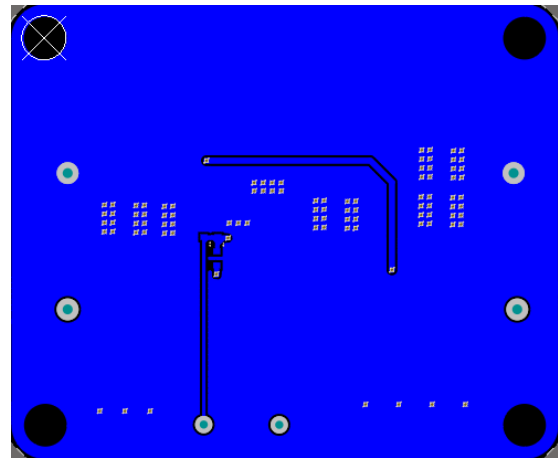


Figure2—Bottom Layer

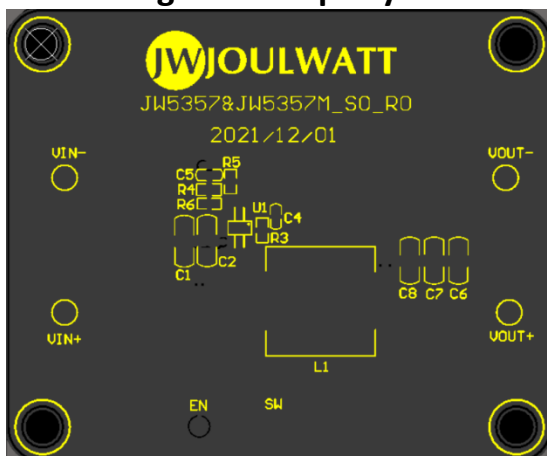


Figure3—Top Silk Layer

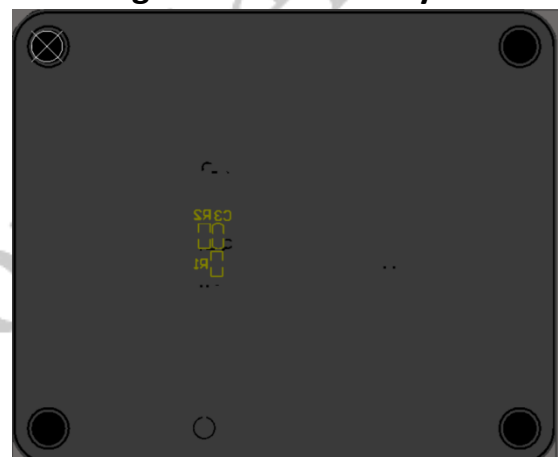


Figure4—Bottom Silk Layer

QUICK START GUIDE

1. Connect the positive terminal and negative terminal of the load to VOUT and GND of EVB, respectively.
2. Connect a power supply between VIN and GND with the supply in “OFF” state. Set the output voltage of the power supply to 4.5V~18V.
3. Turn on the power supply and the evaluation board starts operating in normal condition.
4. The output voltage can be adjusted by varying the R4 and R6 on EVB. For example: Fixed R6 to 11K, when adjusting the output voltage to 3.3V, $R4 = V_{out} / 0.6 * R6 - R6$.
5. For more information, please refer to the datasheet of JW5357&JW5357M.

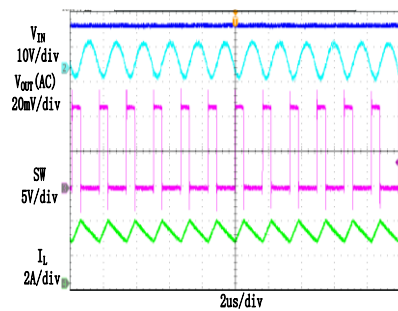
TYPICAL PERFORMANCE CHARACTERISTICS

$V_{IN}=12V$, $V_{OUT}=3.3V$, $L=3.3\mu H$, $C_{OUT}=22\mu F \times 2$, $T_A=+25^\circ C$, unless otherwise noted

Steady State Test

$V_{IN}=12V$, $V_{OUT}=3.3V$

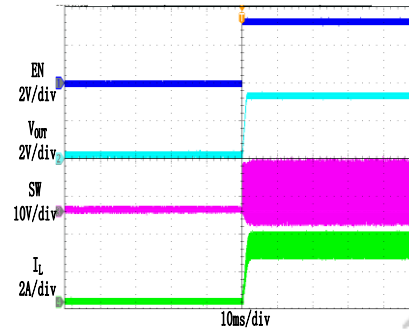
$I_{OUT}=3A$



Startup through Enable

$V_{IN}=12V$, $V_{OUT}=3.3V$

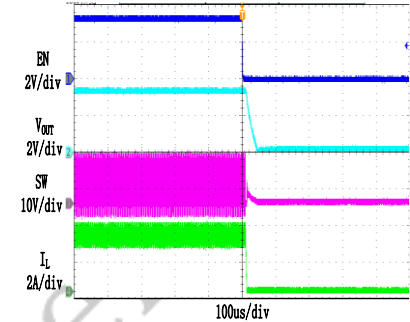
$I_{OUT}=3A$ (Resistive load)



Shutdown through Enable

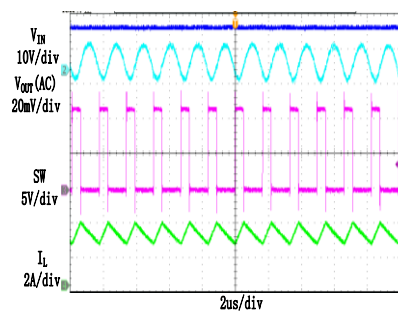
$V_{IN}=12V$, $V_{OUT}=3.3V$

$I_{OUT}=3A$ (Resistive load)



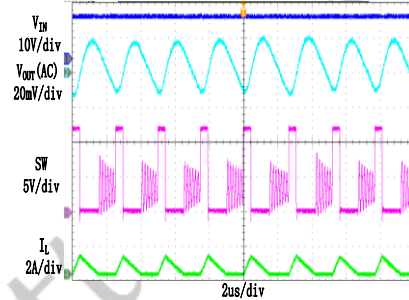
Heavy Load Operation

3A LOAD



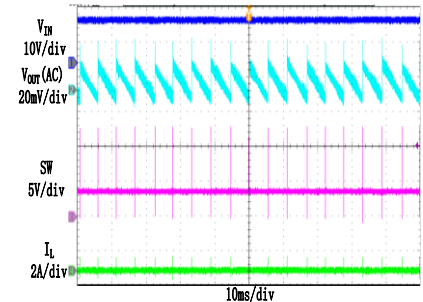
Medium Load Operation

0.3A LOAD



Light Load Operation

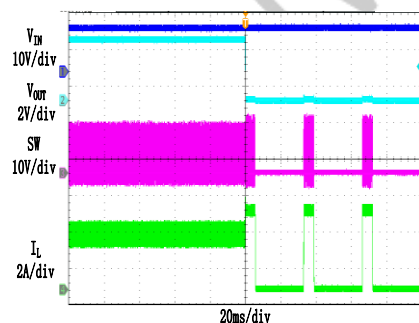
0 A LOAD



Short Circuit Protection

$V_{IN}=12V$, $V_{OUT}=3.3V$

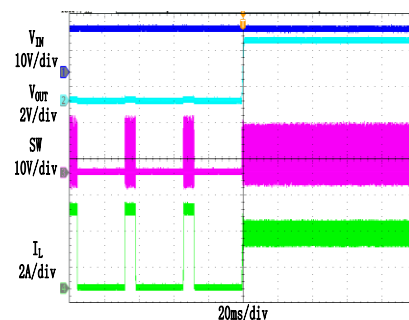
$I_{OUT}=3A$ - Short



Short Circuit Recovery

$V_{IN}=12V$, $V_{OUT}=3.3V$

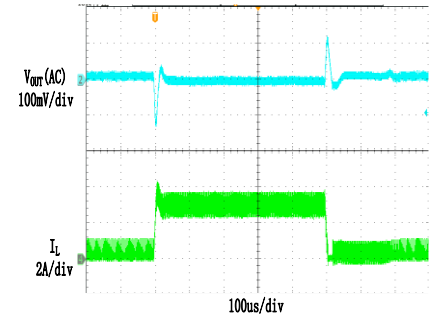
I_{OUT} = Short-3A



Load Transient

$C_5=51pF$

0.3A LOAD $\rightarrow$ 3A LOAD $\rightarrow$ 0.3A LOAD



TYPICAL PERFORMANCE CHARACTERISTICS

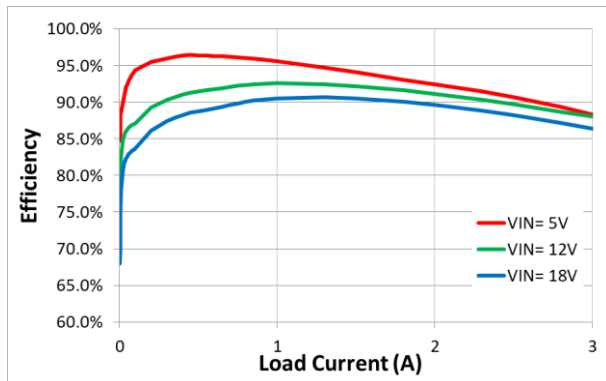


Figure 1. Efficiency vs. Load Current
($V_{OUT}=3.3V$, $L=3.3\mu H$)

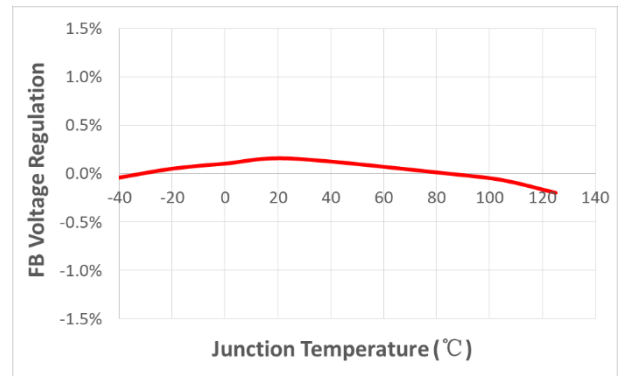


Figure 2. FB Voltage Regulation vs. Junction Temperature

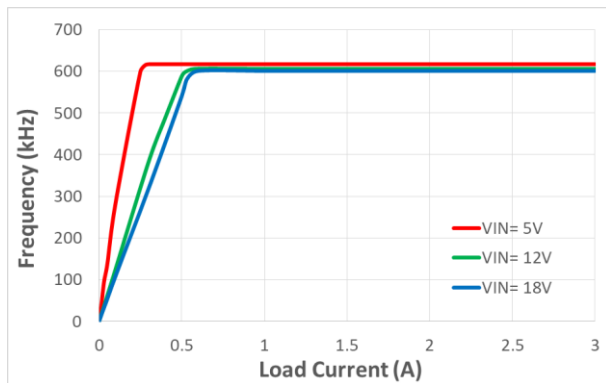


Figure 3. Frequency vs. Load Current
($V_{OUT}=3.3V$, $L=3.3\mu H$)

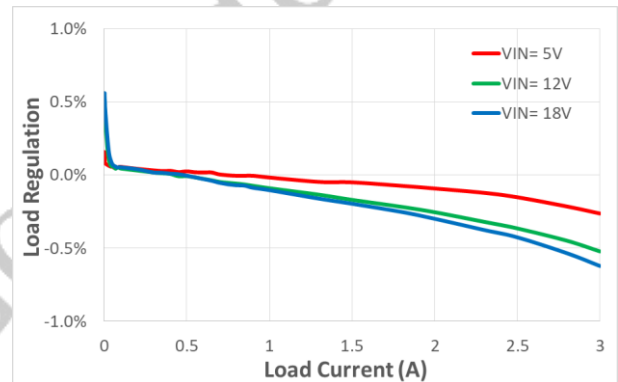


Figure 4. Load Regulation vs. Load Current
($V_{OUT}=3.3V$, $L=3.3\mu H$)

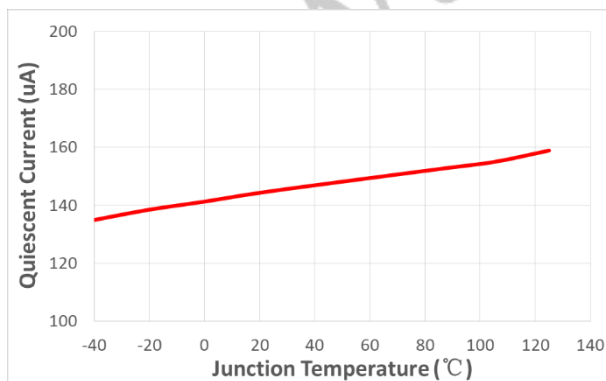


Figure 5. Supply Current vs. Junction Temperature

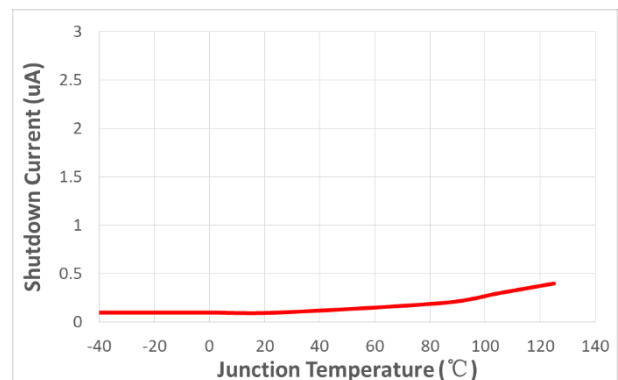


Figure 6. Shutdown Current vs. Junction Temperature

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